

# Patrick Wagstrom

Data Science and Artificial Intelligence Leader

patrick@wagstrom.net, <https://patrick.wagstrom.net/>

Executive leader of cutting edge data science, machine learning, and software engineering organizations. I accelerate value from data by combining best practices from machine learning, risk management, cloud deployment, devops, and human-computer interaction. Hands-on leader that builds cohesive distributed teams by bringing together great people from diverse backgrounds while creating a shared sense of vision and ownership.

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## Professional Experience

### **Chief Data Officer**

May 2021 - December 2022

Brightcove, Boston, MA and Hartford, CT

- Built and managed an organization of 34 FTE and 12 contract engineers, designers, product managers, and scientists responsible for data and model governance, data platforms, analytics and insights, and customer facing data products.
- Executive sponsor and owner of acquisition of Wicket Labs - Brightcove's biggest acquisition in 8 years that brought critical media analytics skills and focus to Brightcove. Brought the product to more than 50x as many customers in under 1 year.
- Ownership of \$Xmm vendor relationship and strategy for a major cloud provider.
- Led team and built out models for machine learning of video classification from text transcripts, audio, and video processing - including development of algorithms and development of model execution platform.
- Responsible for end-to-end analytics collection and processing system for tens of thousands of events a second, hundreds of millions of videos, billions of monthly views, and trillions of rows of data in our data warehouse.

### **Director of Emerging Technology**

September 2019-April 2021

Verizon, Basking Ridge, NJ and Hartford, CT

- Built a team 15 FTEs and additional 51 vendor contractors with a \$15mm annual budget from the ground up - including attracting internal and external talent and negotiating for budget allocations.
- Defined and architect overall enterprise strategy for reproducible machine learning and MLOps including data management, model training, model serving, model refit, and model risk management.
- Led proof-of-concept solutions around blockchain identity and asset management, augmented and virtual reality, cross-device biometrics, IoT, and more.
- Conceived of and built models for cell site energy efficiency resulting in a \$5mm annual savings and conversion into a digital twin for future simulation.
- Led team that developed a holistic model for connecting online brand discussion to customer profiles to support customer servicing - including customer and agent action prediction models and patent pending bot detection model.
- Created and ran numerous cultural and knowledge sharing events including Data Science

Reading Group and Emerging Tech Brown Bag Sessions along with heavy participation in internal and external speaking and outreach events.

### **Senior Director of Data Science for Machine Learning Platforms**

January 2019-September 2019

### **Director of Data Science for Machine Intelligence**

November 2016-January 2019

Capital One, New York, NY and McLean, VA

- Architect and overall lead for the Capital One Card Machine Learning Platform - a platform for scalable and personalized real-time reinforcement machine learning models on top of Kubernetes, gRPC, and AWS inside a regulated industry.
- Led a distributed team of data scientists (12), data engineers (22), and product managers (3) to deliver our platform in a complex cloud-based enterprise environment.
- Led a distributed team to build robust machine learning models that incorporate both unstructured text and structured analytical data to identify customer issues in a large corpus of customer communication.
- Managed, grew, and nurtured a team of 18 data scientists across four states.
- Built out an academic research partnership with universities to explore implications of fairness in artificial intelligence.
- Conducted more than 200 interviews of data scientists, data engineers, and product managers to help grow Card Machine Learning from 33 to 168 people and Capital One's New York Card team from 8 to nearly 200.

### **Research Staff Member/Technical Lead**

January 2015-November 2016

IBM Watson, Littleton, MA

- Led a globally distributed team that built the IBM Watson Conversation service - an innovative service to create rich interactions using natural language processing, entity recognition, and scripted dialog.
- Global team lead for Watson Developer Cloud Tooling. Created cutting edge applications for creating, training, and maintaining cognitive and machine learning solutions including Watson Engagement Advisor and IBM Watson Natural Language Classifier.
- Conducted research on optimal methods to build and configure cognitive and machine learning systems resulting in a 95% reduction in human time required to train these systems.
- Presented cognitive solutions and technologies to more than a dozen companies in six countries.
- Developed best-of-breed modern web application architecture based on Angular, Node.js, GitHub Enterprise, Slack, UrbanCode Deploy and more.

### **Research Staff Member**

January 2014-January 2015

IBM Watson - Watson Life, Yorktown Heights, NY

- Leadership team member responsible for evaluating promising consumer applications of cognitive computing and guiding teams in lean startup and IBM design thinking processes to explore ideas.
- Organized and led globally distributed teams on how to contextualize solutions to cognitive computing challenges.

- Project lead for the IBM Food Truck at SXSW which demonstrated cognitive computing to more than 4,000 people and resulted in more than 1 billion media impressions.
- Planned and executed international workshops on design thinking methodologies.

### **Research Staff Member**

August 2009-January 2014

IBM TJ Watson Research Center, Yorktown Heights, NY

- Analytics lead for JazzHub, IBM's cloud software development strategy. Designed analytics strategy, implemented metrics driven development, introduced A/B testing, and developed analytics dashboards.
- Technical and strategy contributor to IBM's research and acquisition strategy for software development tools and processes.
- Developed and designed GitMiner - an open source project used by 15 universities to perform graph analysis on large scale software engineering databases such as GitHub and BitBucket.
- Led a research team to evaluate productivity of new users and small teams using IBM's enterprise software engineering and product development environments.
- Published papers on topics around distributed collaboration, technical debt in software, and flow of ideas in software engineering communities.

### **Graduate Research Assistant**

August 2003-July 2009

Carnegie Mellon University, Pittsburgh, PA

- Designed and developed CVSMiner an open source tool to perform social network and technical analysis of software engineering ecosystems such as GNOME and Eclipse.
- Worked with members of the GNOME Foundation and Eclipse Foundation to evaluate and improve the relationships between non-profit foundations that manage open source ecosystems and commercial firms.
- Utilized a variety of qualitative and quantitative research methods: stakeholder interviews, message analysis, natural language processing, data mining, machine learning, and social network analysis to generate insight into largely ad hoc software development processes.

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### **Education**

**Ph.D. in Engineering and Public Policy and Computation, Organizations, and Society**, May 2009

Carnegie Mellon University, Pittsburgh, PA

Thesis: "Vertical Interaction in Open Software Engineering Communities". Advisors: Dr. James Herbsleb and Dr. Kathleen Carley.

**MS in Computation, Organizations, and Society**, May 2007

Carnegie Mellon University, Pittsburgh, PA

**MS in Computer Science**, August 2003

Illinois Institute of Technology, Chicago, IL

Thesis: "Scarlet: A Framework for Context Aware Computing". Advisor: Dr. Xian-He Sun.

**BS in Computer Science / BS in Computer Engineering / BS in Electrical Engineering**, May 2002

Illinois Institute of Technology, Chicago, IL